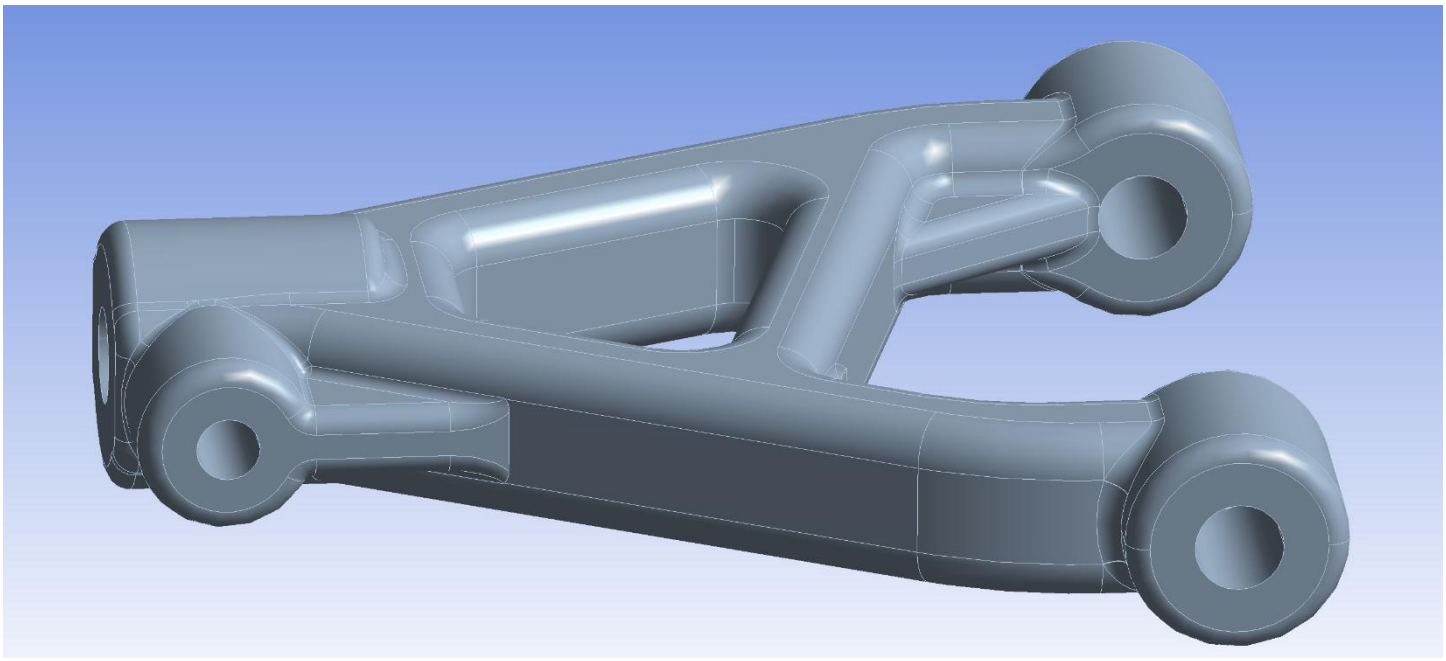
A-Arm Analysis



Contents

- **Units**
- **Model (A4, B4)**
  - Geometry Imports
    - Geometry Import (A3, B3)
  - Geometry
    - Real Front Lower Arm-FreeParts|Scale1
  - Materials
  - Coordinate Systems
  - Mesh
    - Body Sizing
  - Named Selections
  - **Static Structural (A5)**
    - Analysis Settings
    - Loads
    - Solution (A6)
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      - Results
      - Stress Tool
        - Safety Factor
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        - Results
      - Force Reaction
  - **Modal (B5)**
    - Pre-Stress (Static Structural)
    - Analysis Settings
    - Solution (B6)
      - Solution Information
      - Results
- **Material Data**
  - Aluminum alloy, wrought, 6061, T6

Units

TABLE 1

Unit System	Metric (m, kg, N, s, V, A) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A4, B4)

TABLE 2
Model (A4, B4) > Geometry Imports

Object Name	*Geometry Imports*
State	Solved

TABLE 3
Model (A4, B4) > Geometry Imports > Geometry Import (A3, B3)

Object Name	*Geometry Import (A3, B3)*
State	Solved

Definition	
Source	C:\Users\Chaitanya Gajbhiye\AppData\Local\Temp\Ansys - Front Lower Arm.tmp\Ansys - Front Lower Arm_files\import_files\Real Front Lower Arm.STEP
Type	Step
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	No
Parameters	Independent
Parameter Key	ANS;DS
Attributes	No
Attribute Key	SDFEA;DDM
Named Selections	No
Named Selection Key	NS
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	No
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Compare Parts Tolerance	Tight
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	Program Tolerance
Stitch Tolerance	0.0000001
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

Geometry

TABLE 4
Model (A4, B4) > Geometry

Object Name	*Geometry*
State	Fully Defined
Definition	

Source	C:\Users\Chaitanya Gajbhiye\AppData\Local\Temp\Ansys - Front Lower Arm.tmp\Ansys - Front Lower Arm_files\import_files\Real Front Lower Arm.STEP
Type	Step
Length Unit	Inches
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	0.421 m
Length Y	9.525e-002 m
Length Z	0.41402 m
Properties	
Volume	3.6738e-003 m³
Mass	9.9671 kg
Scale Factor Value	1.
Statistics	
Bodies	1
Active Bodies	1
Nodes	683647
Elements	479120
Mesh Metric	None
Update Options	
Assign Default Material	No
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	No
Parameters	Independent
Parameter Key	ANS;DS
Attributes	No
Named Selections	No
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	No
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	Program Tolerance
Decompose Disjoint Geometry	Yes

Enclosure and Symmetry Processing	Yes

TABLE 5
Model (A4, B4) > Geometry > Parts

Object Name	*Real Front Lower Arm-FreeParts|Scale1*
State	Meshed
Graphics Properties	
Visible	Yes
Transparency	1
Definition	
Suppressed	No
Stiffness Behavior	Flexible
Coordinate System	Default Coordinate System
Reference Temperature	By Environment
Treatment	None
Material	
Assignment	Aluminum alloy, wrought, 6061, T6
Nonlinear Effects	Yes
Thermal Strain Effects	Yes
Bounding Box	
Length X	0.421 m
Length Y	9.525e-002 m
Length Z	0.41402 m
Properties	
Volume	3.6738e-003 m³
Mass	9.9671 kg
Centroid X	-1.706e-002 m
Centroid Y	-1.8397e-008 m
Centroid Z	-1.339e-002 m
Moment of Inertia Ip1	0.11886 kg·m²
Moment of Inertia Ip2	0.25797 kg·m²
Moment of Inertia Ip3	0.14668 kg·m²
Statistics	
Nodes	683647
Elements	479120
Mesh Metric	None

TABLE 6
Model (A4, B4) > Materials

Object Name	*Materials*
State	Fully Defined
Statistics	
Materials	2
Material Assignments	0

Coordinate Systems

TABLE 7
Model (A4, B4) > Coordinate Systems > Coordinate System

Object Name	*Global Coordinate System*
State	Fully Defined

Definition	
Type	Cartesian
Coordinate System ID	0.
Origin	
Origin X	0. m
Origin Y	0. m
Origin Z	0. m
Directional Vectors	
X Axis Data	[1. 0. 0.]
Y Axis Data	[0. 1. 0.]
Z Axis Data	[0. 0. 1.]

Mesh

TABLE 8
Model (A4, B4) > Mesh

Object Name	*Mesh*
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Quadratic
Element Size	4.5e-003 m
Sizing	
Use Adaptive Sizing	Yes
Resolution	5
Mesh Defeaturing	Yes
Defeature Size	Default
Transition	Slow
Span Angle Center	Fine
Initial Size Seed	Assembly
Bounding Box Diagonal	0.59811 m
Average Surface Area	1.1125e-003 m²
Minimum Edge Length	2.1747e-005 m
Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Element Quality	Default (5.e-002)
Smoothing	Medium
Mesh Metric	None
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	5
Growth Rate	1.2
Inflation Algorithm	Pre
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No

Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Please Define
Generate Pinch on Refresh	No
Statistics	
Nodes	683647
Elements	479120

TABLE 9
Model (A4, B4) > Mesh > Mesh Controls

Object Name	*Body Sizing*
State	Fully Defined
Scope	
Scoping Method	Geometry Selection
Geometry	1 Body
Definition	
Suppressed	No
Type	Element Size
Element Size	5.e-003 m
Advanced	
Defeature Size	Default
Behavior	Soft

Named Selections

TABLE 10
Model (A4, B4) > Named Selections > Named Selections

Object Name	*Suspension Mount*	*Wheel Mount*	*Chassis Mounts*
State	Fully Defined		
Scope			
Scoping Method	Geometry Selection		
Geometry	2 Faces		4 Faces
Definition			
Send to Solver	Yes		
Protected	Program Controlled		
Visible	Yes		
Program Controlled Inflation	Exclude		
Statistics			
Type	Manual		
Total Selection	2 Faces		4 Faces
Surface Area	4.4968e-003 m²	1.0445e-002 m²	1.5917e-002 m²
Suppressed	0		
Used by Mesh Worksheet	No		

Static Structural (A5)

TABLE 11
Model (A4, B4) > Analysis

Object Name	*Static Structural (A5)*
State	Solved
Definition	

Physics Type	Structural
Analysis Type	Static Structural
Solver Target	Mechanical APDL
Options	
Environment Temperature	20. °C
Generate Input Only	No

TABLE 12
Model (A4, B4) > Static Structural (A5) > Analysis Settings

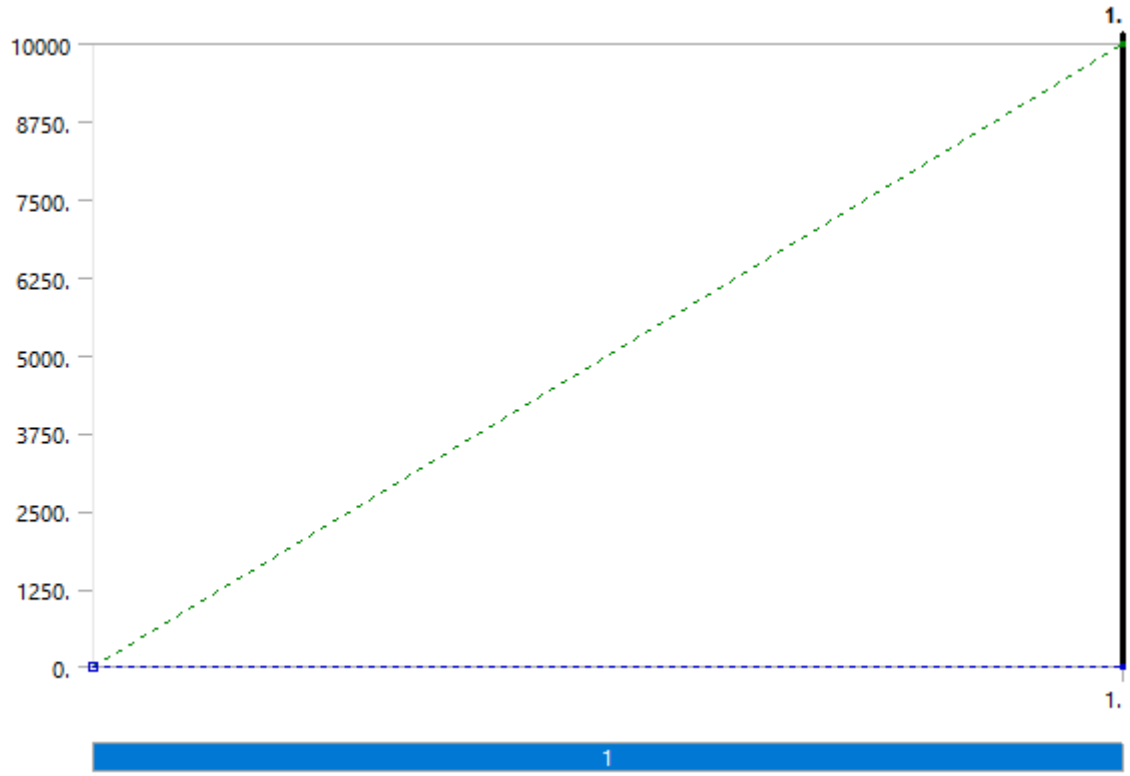
Object Name	*Analysis Settings*
State	Fully Defined
Restart Analysis	
Restart Type	Program Controlled
Status	Done
Step Controls	
Number Of Steps	1.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	Program Controlled
Solver Controls	
Solver Type	Program Controlled
Weak Springs	Off
Solver Pivot Checking	Program Controlled
Large Deflection	On
Inertia Relief	Off
Quasi-Static Solution	Off
Rotordynamics Controls	
Coriolis Effect	Off
Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	Yes
Combine Restart Files	Program Controlled
Nonlinear Controls	
Newton-Raphson Option	Program Controlled
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Program Controlled
Advanced	
Inverse Option	No
Contact Split (DMP)	Off
Output Controls	

Stress	Yes
Surface Stress	No
Back Stress	No
Strain	Yes
Contact Data	Yes
Nonlinear Data	No
Nodal Forces	Yes
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No
Store Results At	All Time Points
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	C:\Users\Chaitanya Gajbhiye\AppData\Local\Temp\Ansys - Front Lower Arm.tmp\Ansys - Front Lower Arm_files\dp0\SYS\MECH\
Future Analysis	Prestressed & StrucOptimization
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	Yes
Solver Units	Active System
Solver Unit System	mks

TABLE 13
Model (A4, B4) > Static Structural (A5) > Loads

Object Name	*Fixed Support*	*Fixed Support 2*	*Force*
State	Fully Defined		
Scope			
Scoping Method	Named Selection		
Named Selection	Chassis Mounts	Suspension Mount	Wheel Mount
Definition			
Type	Fixed Support		Force
Suppressed	No		
Define By			Components
Applied By			Surface Effect
Coordinate System			Global Coordinate System
X Component			0. N (ramped)
Y Component			10000 N (ramped)
Z Component			0. N (ramped)

FIGURE 1
Model (A4, B4) > Static Structural (A5) > Force



Solution (A6)

TABLE 14
Model (A4, B4) > Static Structural (A5) > Solution

Object Name	*Solution (A6)*
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	1 m 39 s
MAPDL Memory Used	4.0264 GB
MAPDL Result File Size	575.75 MB
Post Processing	
Beam Section Results	No
On Demand Stress/Strain	No

TABLE 15
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Solution Information

Object Name	*Solution Information*
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Identify Element Violations	0
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	

Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 16
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Results

Object Name	*Equivalent Stress*	*Total Deformation*
State	Solved	
Scope		
Scoping Method	Geometry Selection	
Geometry	All Bodies	
Definition		
Type	Equivalent (von-Mises) Stress	Total Deformation
By	Time	
Display Time	Last	
Calculate Time History	Yes	
Identifier		
Suppressed	No	
Integration Point Results		
Display Option	Averaged	
Average Across Bodies	No	
Results		
Minimum	65.552 Pa	0. m
Maximum	4.7507e+007 Pa	2.2823e-005 m
Average	1.16e+006 Pa	4.9084e-006 m
Minimum Occurs On	Real Front Lower Arm-FreeParts|Scale1	
Maximum Occurs On	Real Front Lower Arm-FreeParts|Scale1	
Information		
Time	1. s	
Load Step	1	
Substep	1	
Iteration Number	2	

FIGURE 2
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

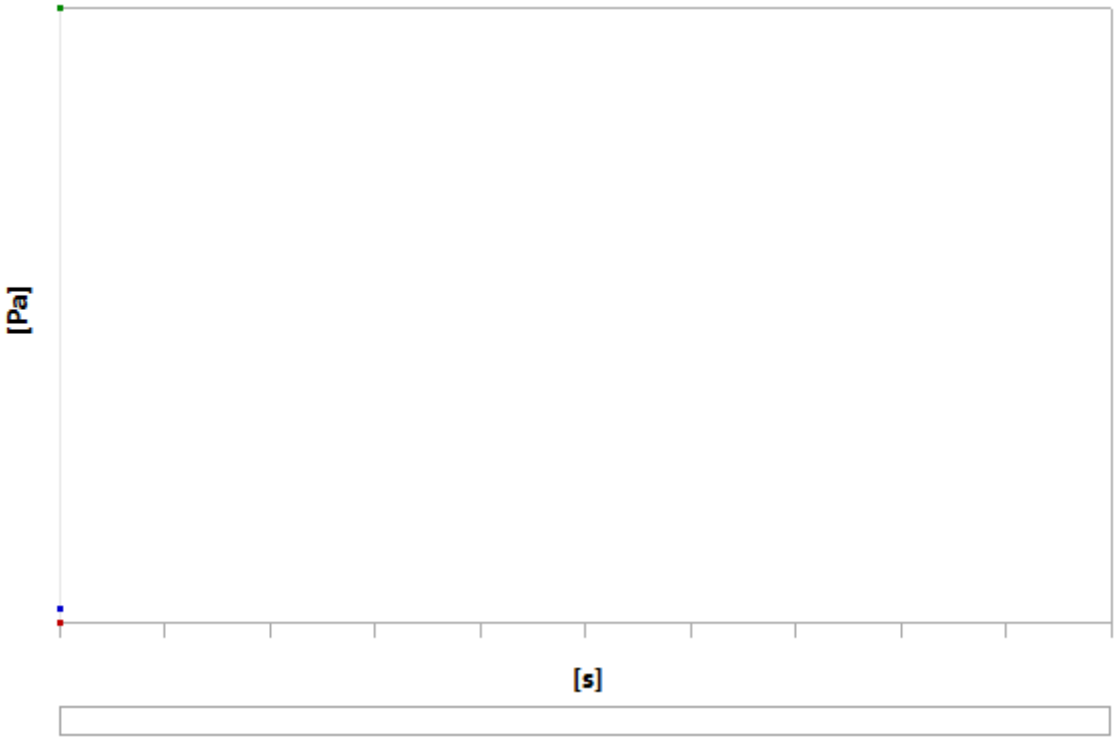


TABLE 17
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

Time [s]	Minimum [Pa]	Maximum [Pa]	Average [Pa]
1.	65.552	4.7507e+007	1.16e+006

FIGURE 3
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Total Deformation

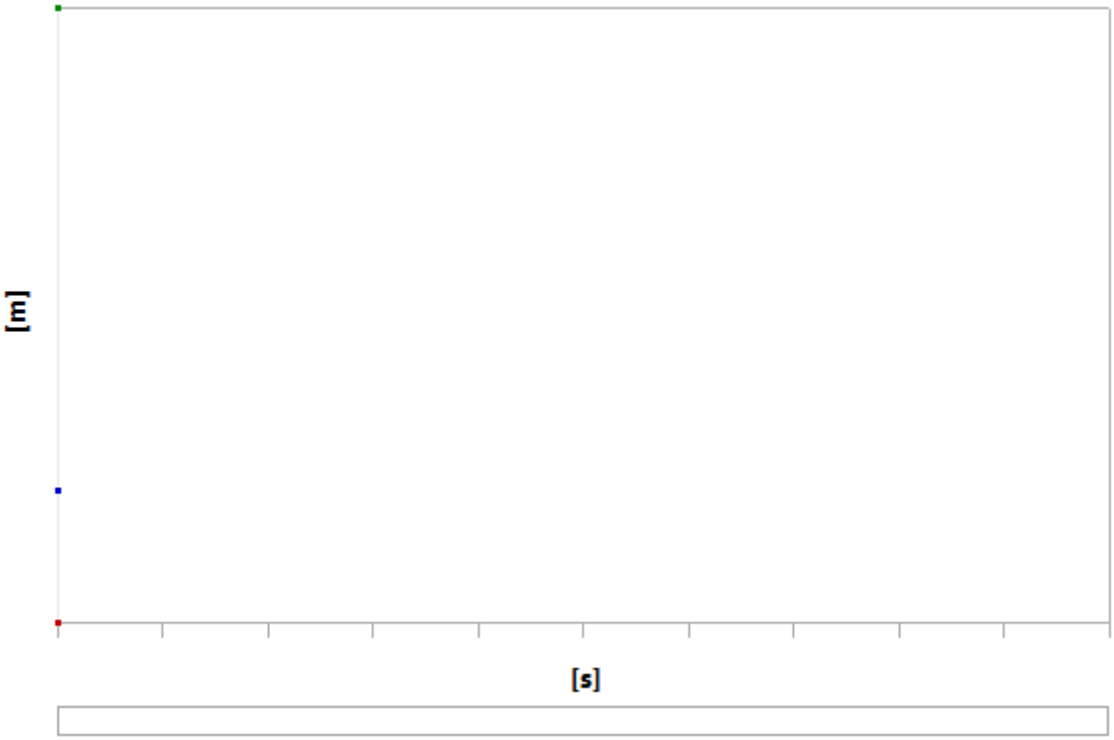


TABLE 18
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Total Deformation

Time [s]	Minimum [m]	Maximum [m]	Average [m]
1.	0.	2.2823e-005	4.9084e-006

TABLE 19
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Stress Safety Tools

Object Name	*Stress Tool*
State	Solved
Definition	
Theory	Max Equivalent Stress
Stress Limit Type	Tensile Yield Per Material

TABLE 20
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Stress Tool > Results

Object Name	*Safety Factor*
State	Solved
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Definition	
Type	Safety Factor
By	Time
Display Time	Last
Calculate Time History	Yes
Identifier	
Suppressed	No
Integration Point Results	
Display Option	Averaged

Average Across Bodies	No
Results	
Minimum	5.456
Minimum Occurs On	Real Front Lower Arm-FreeParts|Scale1
Information	
Time	1. s
Load Step	1
Substep	1
Iteration Number	2

FIGURE 4
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Stress Tool > Safety Factor

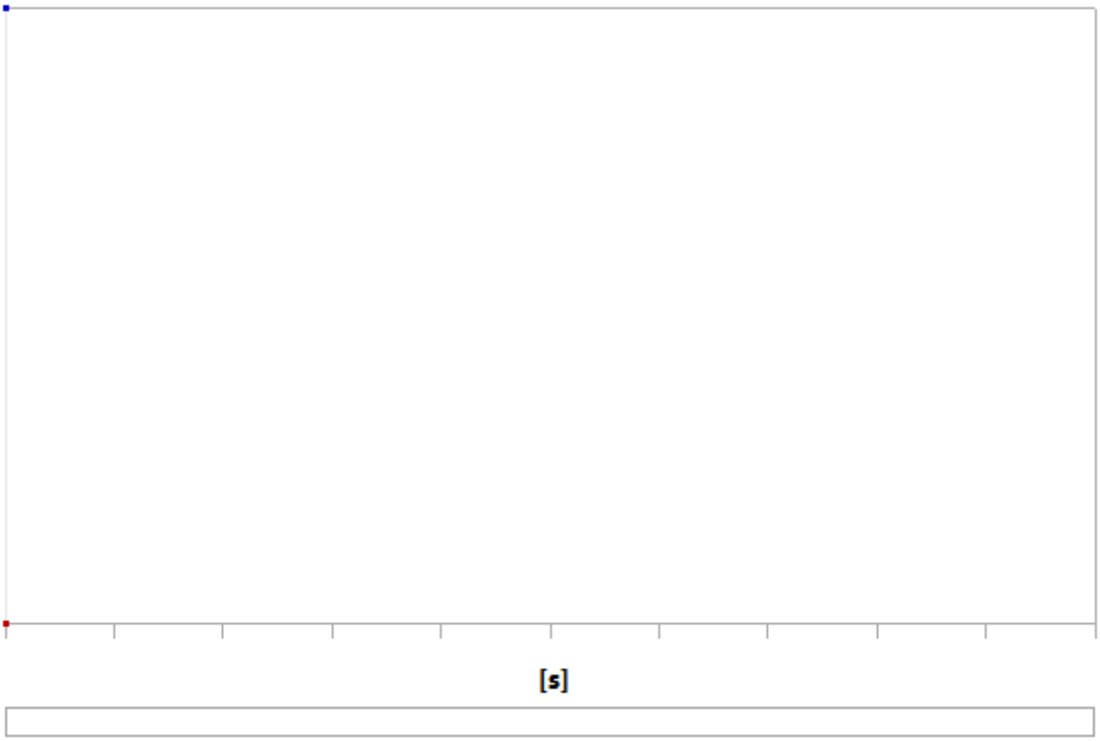


TABLE 21
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Stress Tool > Safety Factor

Time [s]	Minimum	Maximum	Average
1.	5.456	15.	14.997

TABLE 22
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Fatigue Tools

Object Name	*Fatigue Tool*
State	Solved
Domain	
Domain Type	Time
Materials	
Fatigue Strength Factor (Kf)	1.
Loading	
Type	Fully Reversed
Scale Factor	1.
Definition	

Display Time	End Time
Options	
Analysis Type	Stress Life
Mean Stress Theory	None
Stress Component	Equivalent (von-Mises)
Life Units	
Units Name	cycles
1 cycle is equal to	1. cycles

FIGURE 5
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Fatigue Tool

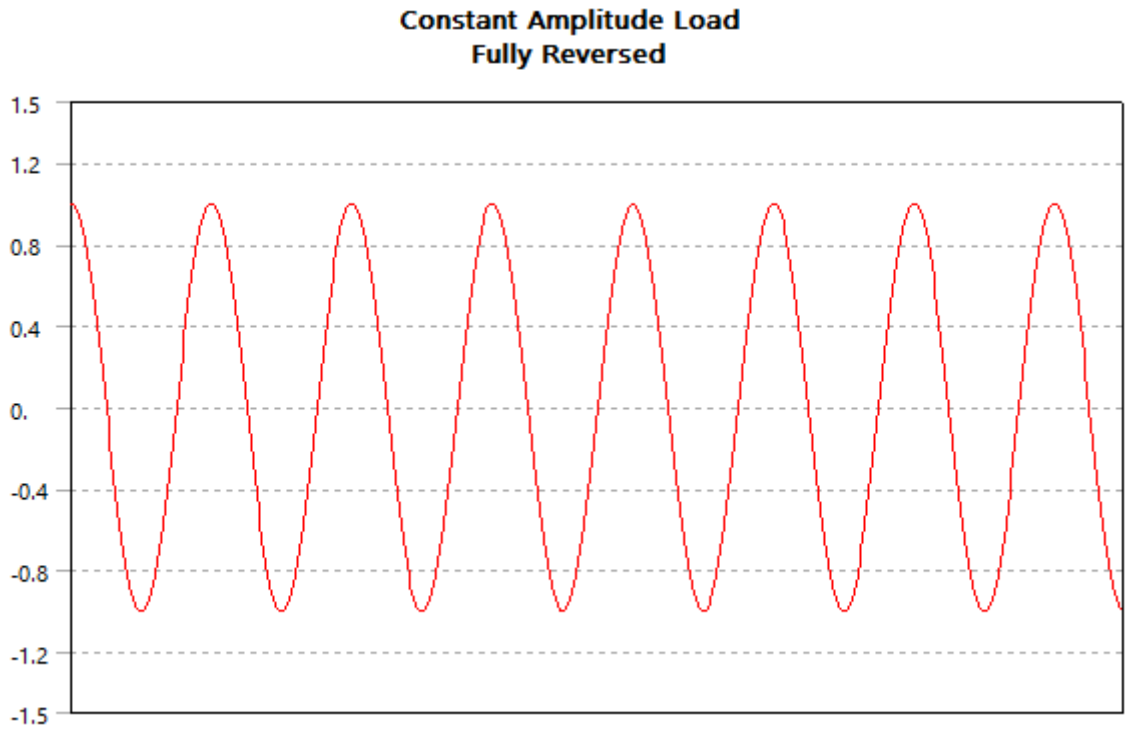


FIGURE 6
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Fatigue Tool

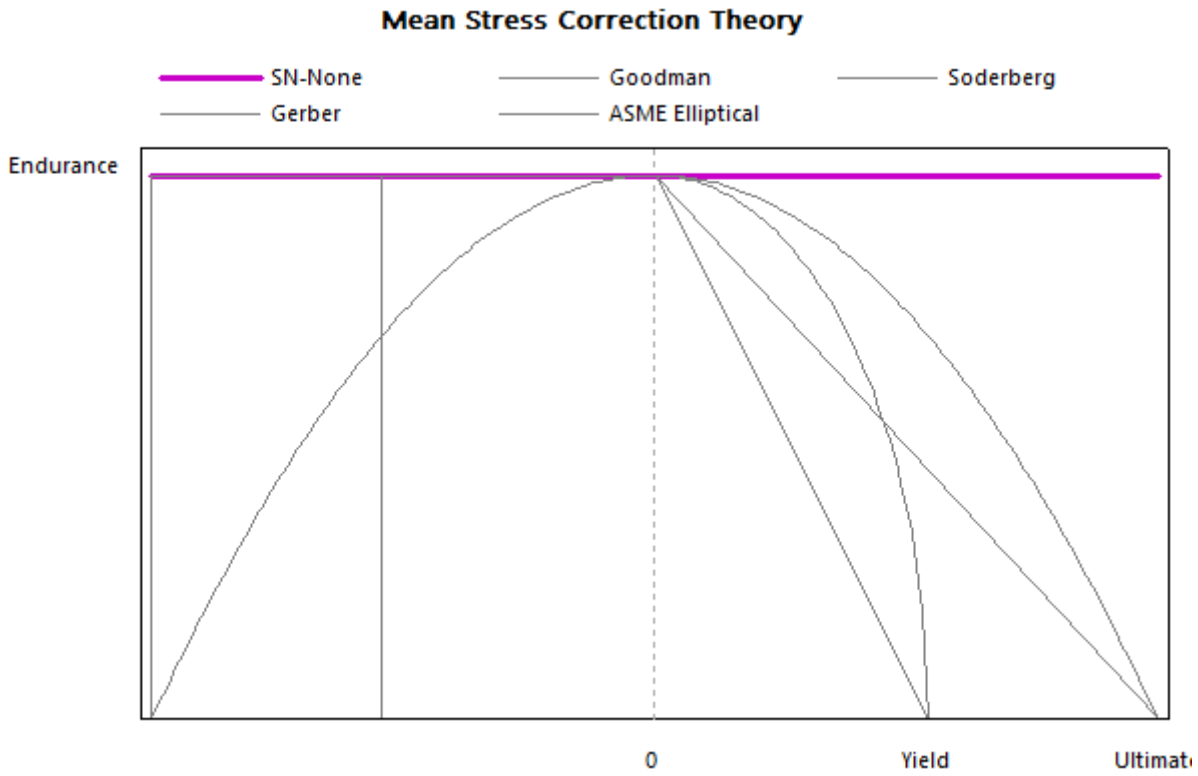


TABLE 23
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Fatigue Tool > Results

Object Name	*Safety Factor*	*Life*	*Damage*
State	Solved		
Scope			
Scoping Method	Geometry Selection		
Geometry	All Bodies		
Definition			
Design Life	1.e+009 cycles		1.e+009 cycles
Type	Safety Factor	Life	Damage
Identifier			
Suppressed	No		
Results			
Minimum	2.2186	1.e+008 cycles	
Minimum Occurs On	Real Front Lower Arm-FreeParts|Scale1		
Maximum			10.
Maximum Occurs On			Real Front Lower Arm-FreeParts|Scale1

FIGURE 7
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Fatigue Tool > Safety Factor

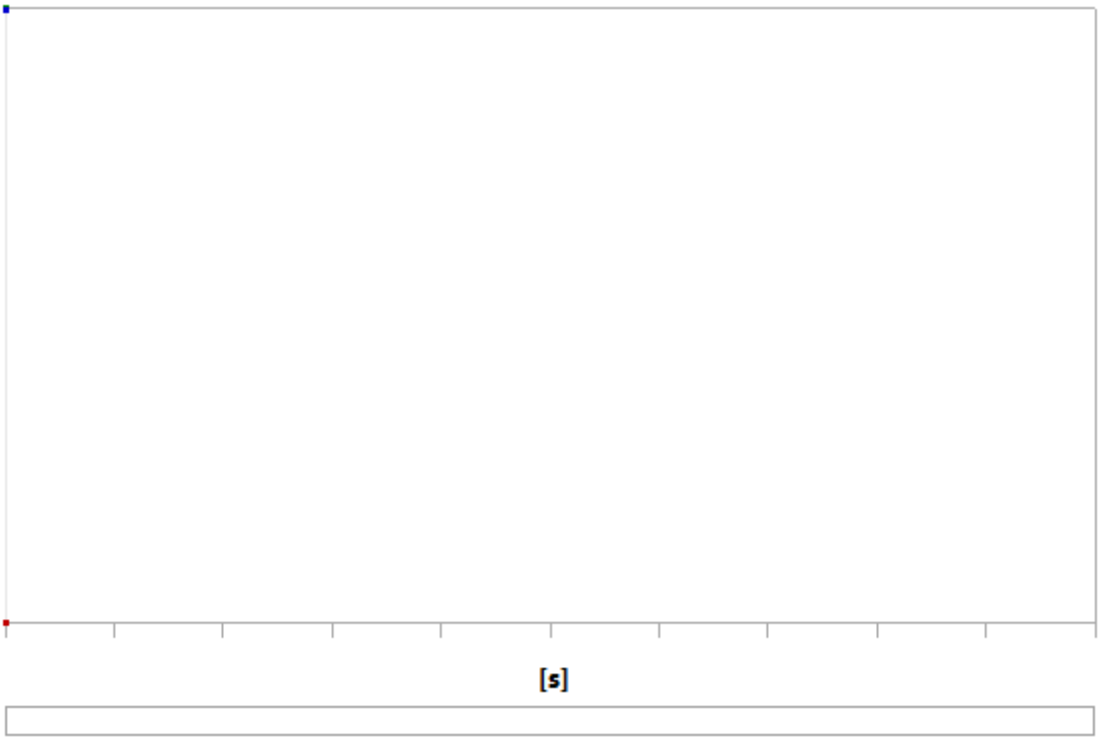


TABLE 24
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Fatigue Tool > Safety Factor

Time [s]	Minimum	Maximum	Average
1.	2.2186	15.	14.952

FIGURE 8
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Fatigue Tool > Life

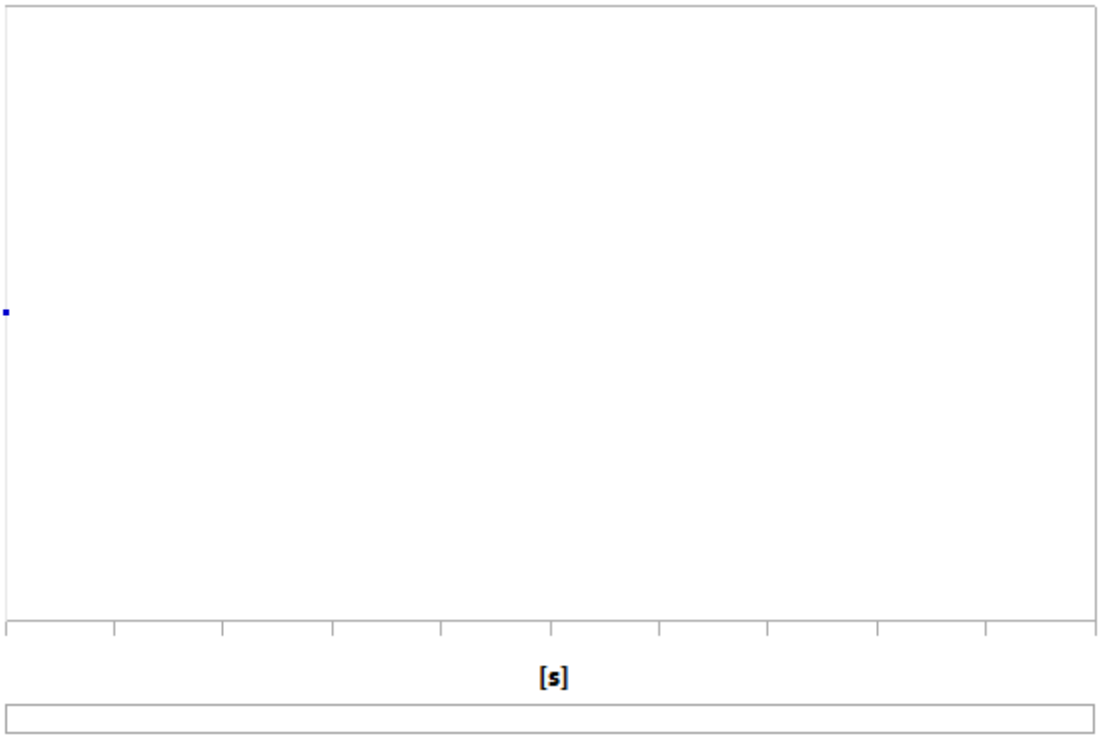


TABLE 25
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Fatigue Tool > Life

Time [s]	Minimum	Maximum	Average
1.	1.e+008	1.e+008	1.e+008

FIGURE 9
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Fatigue Tool > Damage

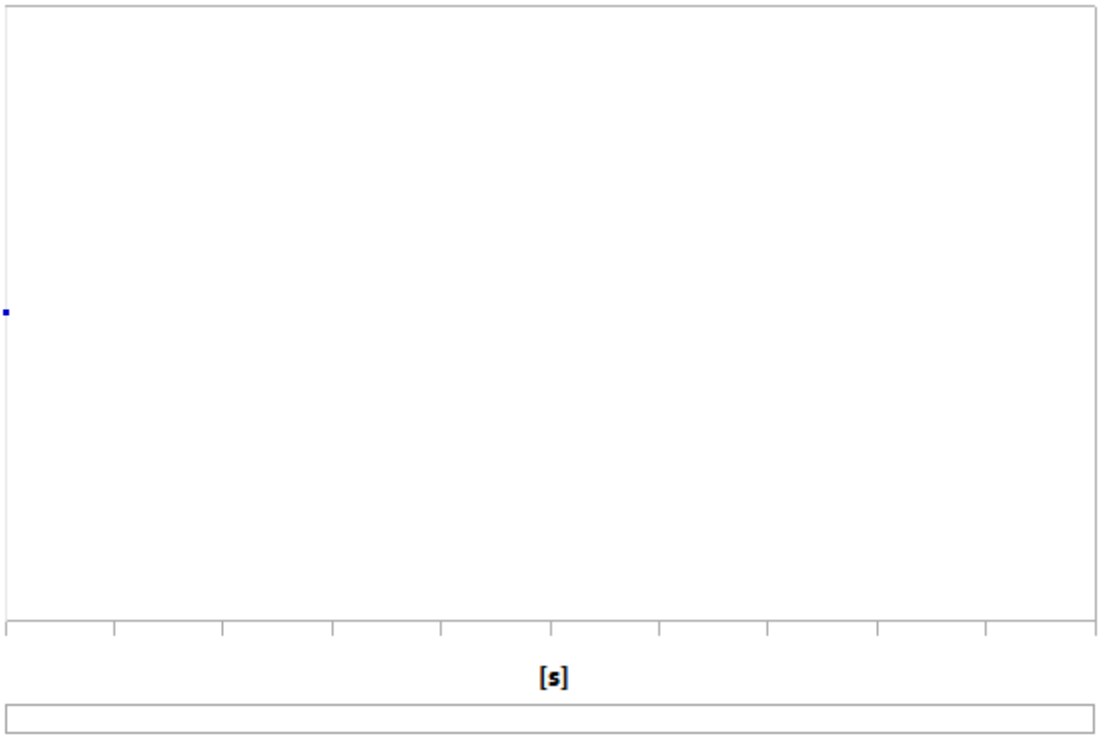


TABLE 26
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Fatigue Tool > Damage

Time [s]	Minimum	Maximum	Average
1.	10.	10.	10.

TABLE 27
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Probes

Object Name	*Force Reaction*
State	Solved
Definition	
Type	Force Reaction
Location Method	Boundary Condition
Boundary Condition	Fixed Support 2
Orientation	Global Coordinate System
Suppressed	No
Options	
Result Selection	All
Display Time	End Time
Results	
X Axis	-0.17104 N
Y Axis	-9467.4 N
Z Axis	0.17515 N
Total	9467.4 N
Maximum Value Over Time	
X Axis	-0.17104 N
Y Axis	-9467.4 N
Z Axis	0.17515 N
Total	9467.4 N
Minimum Value Over Time	

X Axis	-0.17104 N
Y Axis	-9467.4 N
Z Axis	0.17515 N
Total	9467.4 N
Information	
Time	1. s
Load Step	1
Substep	1
Iteration Number	2

FIGURE 10
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Force Reaction

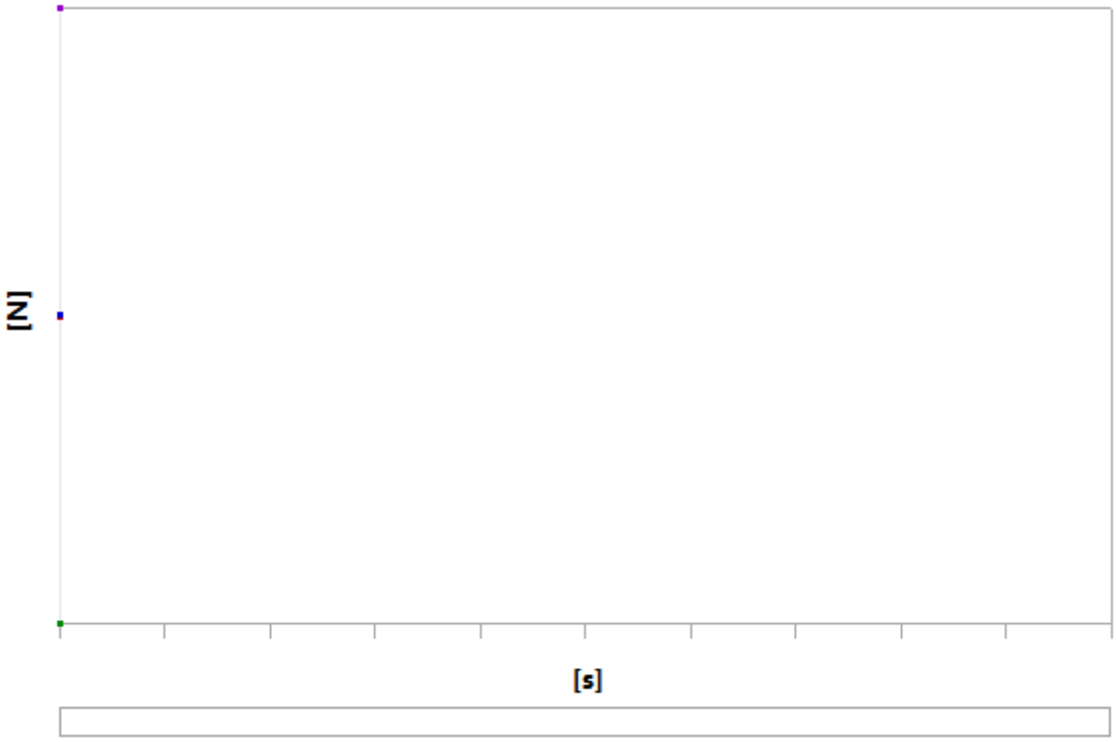


TABLE 28
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Force Reaction

Time [s]	Force Reaction (X) [N]	Force Reaction (Y) [N]	Force Reaction (Z) [N]	Force Reaction (Total) [N]
1.	-0.17104	-9467.4	0.17515	9467.4

Modal (B5)

TABLE 29
Model (A4, B4) > Analysis

Object Name	*Modal (B5)*
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Modal
Solver Target	Mechanical APDL

Options	
Generate Input Only	No

TABLE 30
Model (A4, B4) > Modal (B5) > Initial Condition

Object Name	*Pre-Stress (Static Structural)*
State	Fully Defined
Definition	
Pre-Stress Environment	Static Structural
Pre-Stress Define By	Program Controlled
Reported Loadstep	Last
Reported Substep	Last
Reported Time	End Time
Contact Status	Use True Status
Newton-Raphson Option	Program Controlled

TABLE 31
Model (A4, B4) > Modal (B5) > Analysis Settings

Object Name	*Analysis Settings*
State	Fully Defined
Options	
Max Modes to Find	6
Limit Search to Range	No
On Demand Expansion	No
Spin Softening	Program Controlled
Solver Controls	
Damped	No
Solver Type	Program Controlled
Rotordynamics Controls	
Coriolis Effect	Off
Campbell Diagram	Off
Output Controls	
Stress	No
Surface Stress	No
Back Stress	No
Strain	No
Contact Data	No
Nodal Forces	No
Volume and Energy	No
Euler Angles	No
Calculate Reactions	No
General Miscellaneous	No
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	C:\Users\Chaitanya Gajbhiye\AppData\Local\Temp\Ansys - Front Lower Arm.tmp\Ansys - Front Lower Arm_files\dp0\SYS-1\MECH\
Future Analysis	None
Scratch Solver Files Directory	

Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Solver Units	Active System
Solver Unit System	mks

Solution (B6)

TABLE 32
Model (A4, B4) > Modal (B5) > Solution

Object Name	*Solution (B6)*
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	5 m 9 s
MAPDL Memory Used	6.6973 GB
MAPDL Result File Size	219.44 MB
Post Processing	
Beam Section Results	No

The following bar chart indicates the frequency at each calculated mode.

FIGURE 11
Model (A4, B4) > Modal (B5) > Solution (B6)

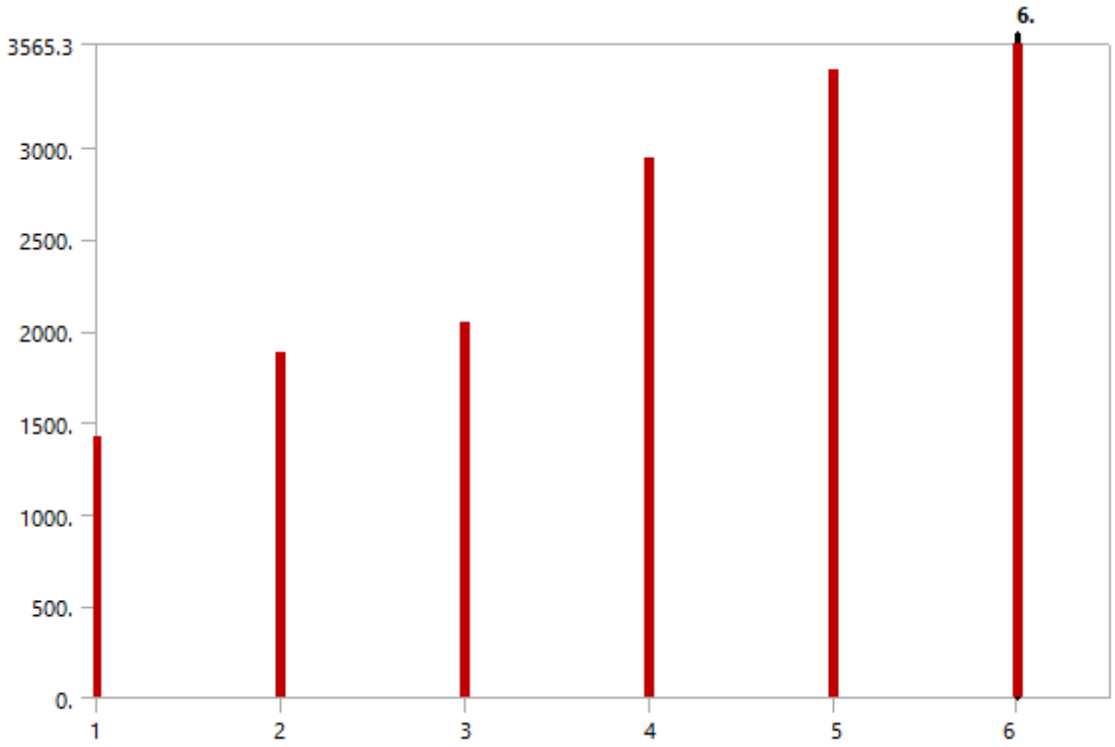


TABLE 33
Model (A4, B4) > Modal (B5) > Solution (B6)

Mode	Frequency [Hz]
1.	1426.
2.	1879.5
3.	2044.8
4.	2946.4
5.	3422.2
6.	3565.3

TABLE 34
Model (A4, B4) > Modal (B5) > Solution (B6) > Solution Information

Object Name	*Solution Information*
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Identify Element Violations	0
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 35
Model (A4, B4) > Modal (B5) > Solution (B6) > Results

Object Name	*Total Deformation*	*Total Deformation 2*	*Total Deformation 3*	*Total Deformation 4*	*Total Deformation 5*	*Total Deformation 6*	*Total Deformation 7*
State	Solved						
Scope							
Scoping Method	Geometry Selection						
Geometry	All Bodies						
Definition							
Type	Total Deformation						
Mode	1.		2.	3.	4.	5.	6.
Identifier							
Suppressed	No						
Results							
Minimum	0. m						
Maximum	0.64263 m		0.68006 m	0.64269 m	0.83978 m	1.0546 m	0.87808 m
Average	0.20053 m		0.20919 m	0.19716 m	0.20839 m	0.14918 m	0.21186 m
Minimum Occurs On	Real Front Lower Arm-FreeParts|Scale1						
Maximum Occurs On	Real Front Lower Arm-FreeParts|Scale1						

Information						
Frequency	1426. Hz	1879.5 Hz	2044.8 Hz	2946.4 Hz	3422.2 Hz	3565.3 Hz

TABLE 36
Model (A4, B4) > Modal (B5) > Solution (B6) > Total Deformation

Mode	Frequency [Hz]
1.	1426.
2.	1879.5
3.	2044.8
4.	2946.4
5.	3422.2
6.	3565.3

TABLE 37
Model (A4, B4) > Modal (B5) > Solution (B6) > Total Deformation 2

Mode	Frequency [Hz]
1.	1426.
2.	1879.5
3.	2044.8
4.	2946.4
5.	3422.2
6.	3565.3

TABLE 38
Model (A4, B4) > Modal (B5) > Solution (B6) > Total Deformation 3

Mode	Frequency [Hz]
1.	1426.
2.	1879.5
3.	2044.8
4.	2946.4
5.	3422.2
6.	3565.3

TABLE 39
Model (A4, B4) > Modal (B5) > Solution (B6) > Total Deformation 4

Mode	Frequency [Hz]
1.	1426.
2.	1879.5
3.	2044.8
4.	2946.4
5.	3422.2
6.	3565.3

TABLE 40
Model (A4, B4) > Modal (B5) > Solution (B6) > Total Deformation 5

Mode	Frequency [Hz]
1.	1426.
2.	1879.5
3.	2044.8
4.	2946.4
5.	3422.2
6.	3565.3

TABLE 41
Model (A4, B4) > Modal (B5) > Solution (B6) > Total Deformation 6

Mode	Frequency [Hz]
1.	1426.
2.	1879.5
3.	2044.8
4.	2946.4
5.	3422.2
6.	3565.3

TABLE 42
Model (A4, B4) > Modal (B5) > Solution (B6) > Total Deformation 7

Mode	Frequency [Hz]
1.	1426.
2.	1879.5
3.	2044.8
4.	2946.4
5.	3422.2
6.	3565.3

Material Data

Aluminum alloy, wrought, 6061, T6

TABLE 43
Aluminum alloy, wrought, 6061, T6 > Constants

Density	2713 kg m^-3
Tensile Yield Strength	2.592e+008 Pa
Tensile Ultimate Strength	3.131e+008 Pa
Resistivity	3.999e-008 ohm m

TABLE 44
Aluminum alloy, wrought, 6061, T6 > Opacity

Red	Green	Blue
234	234	234

Opacity
1

Metallic Finish
1

TABLE 45
Aluminum alloy, wrought, 6061, T6 > Isotropic Elasticity

Young's Modulus Pa	Poisson's Ratio	Bulk Modulus Pa	Shear Modulus Pa	Temperature C
7.66e+010	0.33	7.5098e+010	2.8797e+010	-270.1
7.558e+010	0.33	7.4098e+010	2.8414e+010	-184.6
7.318e+010	0.33	7.1745e+010	2.7511e+010	-99.04
7.029e+010	0.33	6.8912e+010	2.6425e+010	-13.48
6.735e+010	0.33	6.6029e+010	2.532e+010	72.07
6.433e+010	0.33	6.3069e+010	2.4184e+010	157.6
6.078e+010	0.33	5.9588e+010	2.285e+010	243.2
5.579e+010	0.33	5.4696e+010	2.0974e+010	328.7
4.798e+010	0.33	4.7039e+010	1.8038e+010	414.3

3.557e+010	0.33	3.4873e+010	1.3372e+010	499.9

TABLE 46
Aluminum alloy, wrought, 6061, T6 > S-N Curve

Alternating Stress Pa	Cycles	R-Ratio
2.186e+008	1000	-1
2.029e+008	3162	-1
1.885e+008	10000	-1
1.752e+008	31620	-1
1.629e+008	1.e+005	-1
1.515e+008	3.162e+005	-1
1.409e+008	1.e+006	-1
1.31e+008	3.162e+006	-1
1.219e+008	1.e+007	-1
1.134e+008	3.162e+007	-1
1.054e+008	1.e+008	-1
1.286e+008	1000	0
1.229e+008	3162	0
1.175e+008	10000	0
1.122e+008	31620	0
1.07e+008	1.e+005	0
1.02e+008	3.162e+005	0
9.705e+007	1.e+006	0
9.226e+007	3.162e+006	0
8.762e+007	1.e+007	0
8.313e+007	3.162e+007	0
7.878e+007	1.e+008	0
7.055e+007	1000	0.5
6.882e+007	3162	0.5
6.708e+007	10000	0.5
6.532e+007	31620	0.5
6.352e+007	1.e+005	0.5
6.171e+007	3.162e+005	0.5
5.987e+007	1.e+006	0.5
5.801e+007	3.162e+006	0.5
5.614e+007	1.e+007	0.5
5.426e+007	3.162e+007	0.5
5.237e+007	1.e+008	0.5

TABLE 47
Aluminum alloy, wrought, 6061, T6 > Isotropic Secant Coefficient of Thermal Expansion

Coefficient of Thermal Expansion C^-1	Temperature C
1.497e-005	-250.2
1.839e-005	-187.9
2.053e-005	-125.7
2.178e-005	-63.48
2.255e-005	-1.261
2.314e-005	60.96
2.372e-005	123.2
2.435e-005	185.4
2.498e-005	247.6
2.545e-005	309.9

Zero-Thermal-Strain Reference Temperature C	
20	

TABLE 48
Aluminum alloy, wrought, 6061, T6 > Isotropic Thermal Conductivity

Thermal Conductivity W m^-1 C^-1	Temperature C
18.77	-260.2
98.14	-172.4
133.3	-84.59
152.2	3.183
165.8	90.96
175.9	178.7
181.8	266.5
183.6	354.3
181.2	442.1
174.7	529.9

TABLE 49
Aluminum alloy, wrought, 6061, T6 > Specific Heat Constant Pressure

Specific Heat J kg^-1 C^-1	Temperature C
2.414	-260.2
264	-203.7
557.7	-147.3
724.2	-90.86
836.3	-34.44
914.7	21.99
968.5	78.42
1006	134.9
1033	191.3
1055	247.7
1076	304.1
1098	360.6
1123	417
1151	473.4
1179	529.9